

Pavithra Uthrah R. K.

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Profile

Data & AI Engineer with hands-on experience building enterprise-scale data platforms, AI-powered applications, and scalable analytics solutions. Proficient in Python, SQL, PySpark, and cloud platforms (Databricks, Azure) with deep practical exposure to LLMs, RAG, AI Agents, vector search, and MLOps workflows. Experienced across the complete data lifecycle — from ingestion, transformation, and governance to machine learning, Generative AI, and BI consumption. Skilled at translating complex business problems into scalable data and AI solutions, with a track record of delivering production-ready systems. Strong analytical, problem-solving, and cross-functional collaboration skills, demonstrated through internships and leadership of a 250+ member technical community. Passionate about building AI-native data solutions that drive measurable business impact.

Education

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – May 2027

Vellore Institute of Technology, Chennai

Higher Secondary Education (Class XII)

Graduated May 2023

ISC — St. Michael's Academy, Chennai

Technical Skills

- **Data Engineering:** PySpark, Databricks, Delta Lake, ETL/ELT, Medallion Architecture, Dimensional Modeling (Star/Snowflake Schema), Data Warehousing, Apache Airflow, dbt, Unity Catalog, Data Quality, Data Governance, Data Lineage
- **AI & Generative AI:** LLMs, RAG (Retrieval-Augmented Generation), Vector Search, Embeddings, AI Agents, Prompt Engineering, LangChain, LangGraph, Databricks Genie, Mosaic AI, Model Serving, OpenAI/Gemini APIs
- **Machine Learning & Statistics:** Regression, Classification, XGBoost, Decision Trees, Random Forest, Clustering, Time Series Forecasting (ARIMA, Prophet), Feature Engineering, Model Evaluation (ROC-AUC, RMSE, MAE), SHAP Explainability, Statistical Analysis, Hypothesis Testing, EDA
- **Programming & Libraries:** Python (Pandas, NumPy, Scikit-learn), SQL, Java, PySpark, Matplotlib, Seaborn
- **Cloud Platforms:** Microsoft Azure (Data Factory, Synapse Analytics, ADLS Gen2), Databricks, Snowflake, Google BigQuery
- **Databases & Storage:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Elasticsearch
- **Visualization & BI:** Power BI, Tableau, Power Query, DAX
- **DevOps & Tools:** Git/GitHub, Docker, Azure DevOps, Jenkins, CI/CD, Linux/Unix

Work Experience

Data Engineer – Kaar Technologies, Chennai

Internship | On-site | Jul 2026 – Present

- Built an enterprise-scale EV Fleet Lakehouse Platform processing 50GB+ of vehicle telemetry through Medallion Architecture using Databricks, PySpark, and Delta Lake — enabling real-time operational analytics for fleet utilization, battery efficiency, and maintenance.
- Designed an AI-Powered SAP ERP Intelligence Assistant using Databricks Genie, RAG, Vector Search, and LLMs — enabling natural-language access to enterprise data and knowledge.
- Architected an AI-Powered Multi-Agent Data Engineering Assistant combining Data Quality, SLA Monitoring, Root Cause Analysis, Predictive Failure Detection, and Self-Healing agents using LangGraph and Databricks AI/Genie.
- Implemented data quality, lineage tracking, and governance using Unity Catalog — ensuring reliable, governed data for downstream analytics and AI.
- Delivered Power BI dashboards for operational KPIs, translating complex telemetry data into actionable insights for stakeholders.
- Collaborated with 6 cross-functional stakeholders to translate business requirements into scalable data and AI solutions.

Software Development Engineer – AIC Crescent Innovation & Incubation Council, Chennai

Internship | On-site | May 2026 – Jun 2026

- Architected and built a multi-tenant ERP backend skeleton supporting 14 entities and 140+ startup organizations using Django REST Framework, PostgreSQL, and RBAC.
- Developed an AI-powered WhatsApp workflow automation prototype using Python, REST APIs, and OpenAI/Gemini APIs — extracting structured tasks from unstructured messages and reducing manual reporting effort by 75%.
- Applied prompt engineering and AI-assisted development for rapid prototyping, debugging, and documentation.
- Participated in code reviews and unit test creation, contributing to software quality and maintainability.
- Wrote comprehensive technical documentation and architecture notes to enable smooth handoff.

Projects

AI-Powered Multi-Agent Data Engineering Assistant

LangGraph, Databricks AI/Genie, Unity Catalog, Model Serving, LLMs

- Built an AI-powered multi-agent system on Databricks enabling users to monitor, diagnose, predict, and resolve data pipeline issues through natural language.
- Architected a coordinated agent framework combining Data Quality, SLA Monitoring, Root Cause Analysis, Predictive Failure Detection, and What-If Simulation agents.
- Implemented Self-Healing agent using tool/function calling, multi-step workflows, and connecting LLMs with external systems and APIs.

AI-Powered SAP ERP Intelligence Assistant

Databricks, Genie, RAG, Vector Search, LLM, Mosaic AI, LangChain

- Conceptualized and designed a full-scale AI system that masks SAP complexity for business users through a unified natural-language interface to enterprise ERP data and knowledge.
- Architected modular pipeline: Structured SAP data queried via Databricks Genie, while unstructured ERP documents processed through RAG/Vector Search pipeline using embedding models.
- Designed intelligent query router that classifies user intent (analytics/knowledge/hybrid) and orchestrates blended responses unified by an LLM.

AI-Driven Automated ETL Pipeline

Databricks Lakeflow, SQL AI Functions, Delta Lake, Serverless

- Architected an AI-native automated ETL pipeline using Databricks Lakeflow and SQL AI Functions — eliminating traditional ETL tools and manual Python parsing scripts.
- Applied SQL AI Functions (`ai_analyze_sentiment`, `ai_classify`, `ai_extract`) for AI-powered data enrichment, converting unstructured text into analytics-ready structured attributes.
- Designed serverless architecture removing manual infrastructure provisioning and management.

Enterprise EV Fleet Lakehouse Platform

Databricks, PySpark, Delta Lake, Unity Catalog, dbt, Airflow, Power BI

- Engineered an enterprise-scale Lakehouse processing 50GB+ of vehicle telemetry through Medallion Architecture using Databricks, PySpark, and Delta Lake.
- Implemented incremental ingestion with ACID transactions, schema evolution, and time travel; configured Unity Catalog for governance, metadata management, and lineage tracking.
- Developed automated data quality validation and optimized Spark jobs, reducing pipeline execution time by 40% and compute costs by 30%.
- Delivered Power BI dashboards for fleet utilization, battery efficiency, and operational KPIs.

Customer Churn Prediction & Retention Strategy Model

Python, Scikit-learn, XGBoost, SHAP, Pandas, NumPy, Power BI

- Built a customer churn prediction model on 85K+ customer records using XGBoost (ROC-AUC: 0.75) to identify at-risk customers for proactive retention.
- Engineered 15+ behavioral features through extensive EDA and feature engineering, identifying key drivers of churn.
- Applied SHAP for model explainability to identify top churn drivers, translating technical insights into 3 actionable business recommendations.
- Built a Power BI dashboard visualizing churn-risk segments and model outputs for stakeholder decision-making.

Real-Time IoT Sensor Streaming & Anomaly Detection Pipeline

Kafka, Spark Structured Streaming, PostgreSQL, Docker, Python

- Engineered a streaming pipeline processing 2,500 simulated IoT sensor events/second through Kafka, with Spark Structured Streaming for real-time aggregation and anomaly detection.
- Implemented windowed aggregations and late-data handling, achieving 99.2% processing accuracy under simulated delays.
- Containerized the full stack with Docker Compose for reproducible deployment.

Certifications

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| • Google Data Analytics Professional Certificate (v.3) | Issued by Google, January 2026. |
| • Data Science & Analytics Certificate | Issued by HP LIFE, August 2026. |

Leadership & Activities

- Chairperson, Open Source Programming Club, VIT Chennai (Apr 2025 – Present)
 - Lead a 250+ member technical community, driving strategy, operations, partnerships, and cross-functional initiatives. Organized 3 national-level hackathons and 10+ technical events, workshops, coding competitions, and speaker sessions.

- Scaled the club to become the university's No. 1 technical club within two years of its launch, growing membership to 300+ and organizing 30+ national technical events.
- **Student Mentor Volunteer, U&I Trust, India** (Jul 2026 – Present)
 - Provide weekly academic tutoring and holistic mentorship to primary school students, supporting their educational progress, confidence, life skills, and personal development through consistent guidance and positive role modelling.
- **Organizer – Spectrum, VOID.v1, Glytch, TechnoVIT'25, VIT Chennai**
 - Coordinated operations, participant engagement, and logistics for university technical festivals and hackathons involving 3000+ participants; served on the Organizing Committee for 10+ other technical events.
- **Rotaractor, Rotaract Club of VIT Chennai** (Jul 2025 – Jun 2026)
- **Management Member, Microsoft Innovations Club, VIT Chennai** (Sep 2024 – Apr 2025)